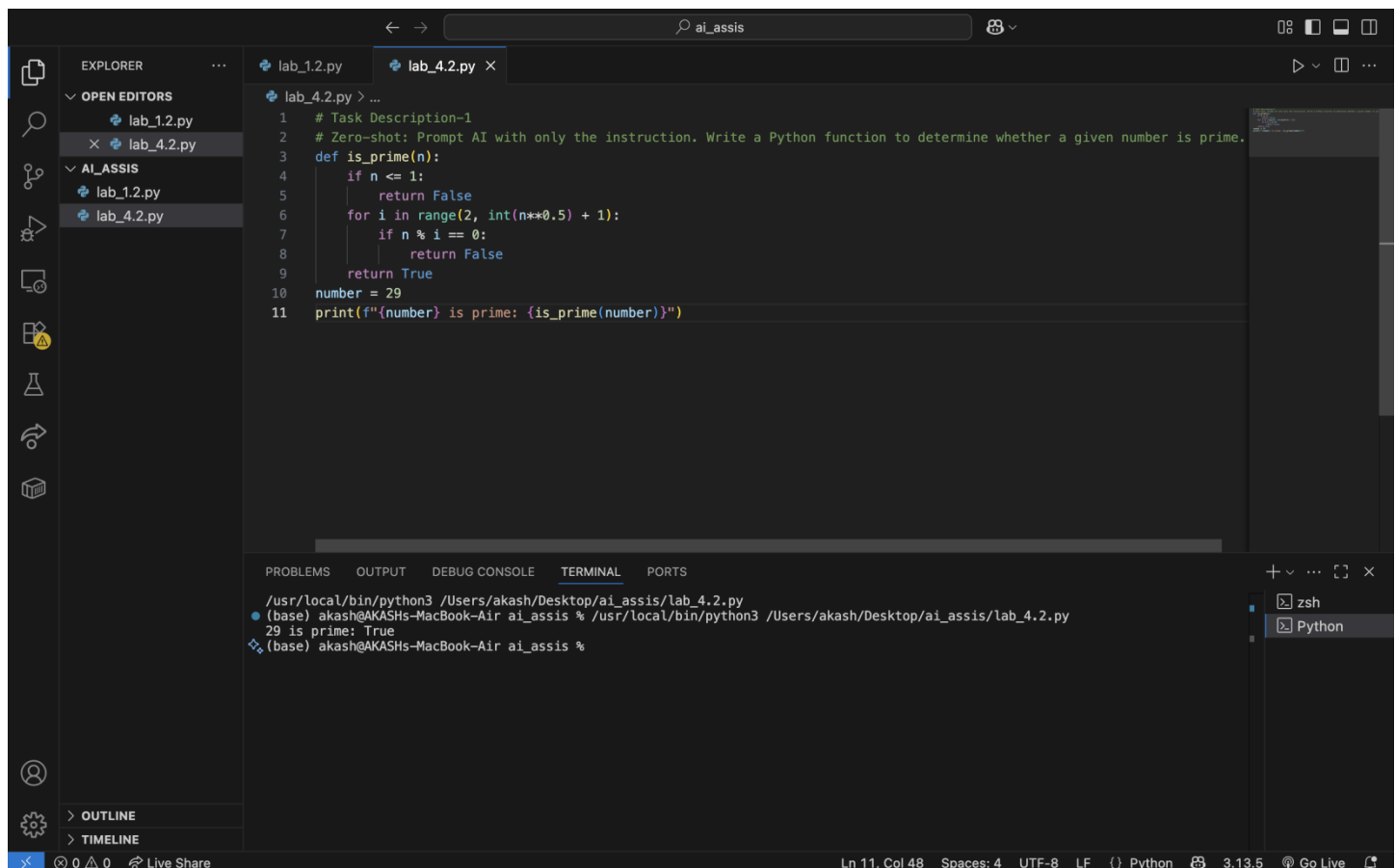


School of Computer Science and Artificial Intelligence**Lab Assignment # 4.2**

Program : B. Tech (CSE)
Specialization : --
Course Title : AI Assisted coding
Course Code :
Semester II
Academic Session : 2025-2026
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Batch No. : 52
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Task Description-1

Zero-shot: Prompt AI with only the instruction. Write a Python function to determine whether a given number is prime.



The screenshot shows a VS Code editor with a file named `lab_4.2.py` open. The code defines a function `is_prime(n)` that checks if a number `n` is prime by testing divisibility from 2 to `int(n**0.5) + 1`. It then sets `number = 29` and prints the result of `is_prime(number)`. The terminal output shows the command `/usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py` being executed, resulting in the output `29 is prime: True`.

```
1 # Task Description-1
2 # Zero-shot: Prompt AI with only the instruction. Write a Python function to determine whether a given number is prime.
3 def is_prime(n):
4     if n <= 1:
5         return False
6     for i in range(2, int(n**0.5) + 1):
7         if n % i == 0:
8             return False
9     return True
10 number = 29
11 print(f"{number} is prime: {is_prime(number)}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
/usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py
(base) akash@AKASHs-MacBook-Air ai_assis % /usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py
29 is prime: True
(base) akash@AKASHs-MacBook-Air ai_assis %
```

Explanation:

The function `is_prime` checks if a number `n` is prime by testing divisibility from 2 to the square root of `n`.
If any divisor is found, it returns False; otherwise, it returns True.
Few-shot: Provide AI with examples before the instruction.
Example 1: 5 is prime because it has no divisors other than 1 and itself.

Task Description-2

- One-shot: Provide one example: Input: [1, 2, 3, 4], Output: 10 to help AI generate a function that calculates the sum of elements in a list.

```
11 # print(' {number} is prime: {is_prime(number)} ')
```

```
12
```

```
13 # Task Description-2
```

```
14 # One-shot: Provide one example: Input: [1, 2, 3, 4],
```

```
15 # Output: 10 to help AI generate a function that calculates the sum of elements in a list.
```

```
16 def sum_of_list(lst):
```

```
17     | return sum(lst)
```

```
18 numbers = [1, 2, 3, 4]
```

```
19 print("Sum of list:", sum_of_list(numbers))
```

```
20
```

```
21
```

```
22
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

/usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py

• (base) akash@AKASHs-MacBook-Air ai_assis % /usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py

Sum of list: 10

❖ (base) akash@AKASHs-MacBook-Air ai_assis %

Explanation

```
# The function sum_of_list takes a list as input and returns the sum of its elements using Python's built-in sum() function.
```

Task Description-3

- Few-shot: Give 2–3 examples to create a function that extracts digits from an alphanumeric string.

```

20
21 # Task Description-3
22 # Few-shot: Give 2-3 examples to create a function that extracts digits from an alphanumeric string.
23 def extract_digits(s):
24     return ''.join(filter(str.isdigit, s))
25
26 # Example 1
27 input_str1 = "abc123def456"
28 output1 = extract_digits(input_str1)
29 print("Extracted digits:", output1)
30
31 # Example 2
32 input_str2 = "no_digits_here!"
33 output2 = extract_digits(input_str2)
34 print("Extracted digits:", output2)
35
36 # Example 3
37 input_str3 = "2024year!"
38 output3 = extract_digits(input_str3)
39 print("Extracted digits:", output3)
40

```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Python

```

/usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py
(base) akash@AKASHs-MacBook-Air ai_assis % /usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py
Extracted digits: 123456
Extracted digits:
Extracted digits: 2024
(base) akash@AKASHs-MacBook-Air ai_assis %

```

Explanation

```
# The function `extract_digits` takes an alphanumeric string as input and uses the `filter` function along with `
# str.isdigit` to extract only the digit characters from the string.
# It then joins these characters into a single string and returns it.
```

The examples demonstrate the function's ability to extract digits from different types of input strings.

Task Description-4

- Compare zero-shot vs few-shot prompting for generating a function that counts the number of vowels in a string.

```
40
41
42 # Task Description-4
43 # [ ] Compare zero-shot vs few-shot prompting for generating a function that counts the number of vowels in a string.
44 def count_vowels(s):
45     vowels = "aeiouAEIOU"
46     return sum(1 for char in s if char in vowels)
47
48 # Zero-shot example
49 input_str = "Hello World!"
50 print("Number of vowels (zero-shot):", count_vowels(input_str))
51
52 # Few-shot examples
53 input_str1 = "This is a test."
54 input_str2 = "Another example."
55 print("Number of vowels (few-shot):", count_vowels(input_str1))
56 print("Number of vowels (few-shot):", count_vowels(input_str2))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

/usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py
● (base) akash@AKASHs-MacBook-Air ai_assis % /usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py
Number of vowels (zero-shot): 3
Number of vowels (few-shot): 4
Number of vowels (few-shot): 6
❖ (base) akash@AKASHs-MacBook-Air ai_assis %

Comparison

The zero-shot approach relies solely on the initial instruction, which may lead to less accurate or less optimized code.

The few-shot approach, by providing examples, helps the AI understand the context better and often results in more accurate and efficient implementations. In this case, both approaches yield the same function, but few-shot prompting can enhance understanding for more complex tasks.

Task Description-5

Use few-shot prompting with 3 sample inputs to generate a function that determines the minimum of three numbers without using

```
59
60 # Task Description-5
61 # [ ] Use few-shot prompting with 3 sample inputs to generate a function that determines the minimum of three numbers
62 # without using the built-in min() function.
63 def find_minimum(a, b, c):
64     if a < b and a < c:
65         return a
66     elif b < c:
67         return b
68     else:
69         return c
70
71 # Example inputs
72 print("Minimum (few-shot):", find_minimum(3, 1, 2))
73 print("Minimum (few-shot):", find_minimum(5, 10, 7))
74 print("Minimum (few-shot):", find_minimum(-1, -5, -3))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

/usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py
● (base) akash@AKASHs-MacBook-Air ai_assis % /usr/local/bin/python3 /Users/akash/Desktop/ai_assis/lab_4.2.py
Minimum (few-shot): 1
Minimum (few-shot): 5
Minimum (few-shot): -5
❖ (base) akash@AKASHs-MacBook-Air ai_assis %

Explanation

```
# The function find_minimum takes three numbers as input and compares them using conditional statements to determine the smallest number.
```